ENVS 193DS Final

Steven Le

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Homework set up

```
#read in packages
library(tidyverse)
library(here)
library(janitor)
library(readxl)
library(ggplot2)
library(ggeffects)
library(flextable)
library(here)
library(MuMIn)
library(DHARMa)
library(rstatix)
#stored nest data as an object called nests
nests <- read_csv(here("data", "nest_data_final.csv"))
#stored personal data as an object called personal
personal <- read_csv(here("data", "personal_data.csv"))
```

Problem 1. Research writing

a. Transparent statistical methods

In part 1, they used a logistic regression, which is a generalized linear model with a binomial error distribution. In part 2, they used a chi-square test, which is used to identify if there is a relationship between two categorical variables.

b. Suggestions for rewriting

Part 1.

The distance to water edge does seem to predict the probability of American Avocet microhabitat use.

Part 2.

There does seem to be a relationship between bird species (American Avocet, Forster's Tern, Black-necked Stilt) and habitat (managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat).

c. Further analysis

One additional variable that could influence the probability of American Avocet microhabitat use is the type of behavior they were displaying, which is a categorical variable. For the American Avocets, the researchers put behavior into four main categories: feeding (searching, probing, and dunking bills into the water), roosting (standing, sitting, preening), breeding (nest-building, incubating, brooding chicks, alarm calling, breeding display, copulation, or predator distraction display), and other (walking, swimming, or flushing). This variable might influence American Avocet habitat use as different behaviors can change the duration of how long they remain in a particular habitat.

Problem 2. Data analysis

a. Clean and wrangle

```
#created new object from nests
nests_clean <- nests |>
#cleaned column names
  clean_names() |>
#selected for tree height, ant_species, and case_control columns
  select(case_control, ant_species, height_cm) |>
#reordered names in ant_species column
  mutate(ant_species = as_factor(ant_species),
         ant_species = fct_relevel(ant_species,
                                   "AB",
                                   "RRB",
                                   "BBR")) |>
#filtered ant_species column to only have three required species
  filter(ant_species %in% c("AB", "RRB", "BBR")) |>
#created new column with full scientific name for each ant species
  mutate(scientific_name = case_match(ant_species,
    "AB" ~ "Crematogaster sjostedti",
    "RRB" ~ "Crematogaster mimosae",
```

```
    "BBR" ~ "Crematogaster nigriceps")) |>
#reordered full scientific name of ant species
  mutate(scientific_name = factor(scientific_name,
                                levels = c("Crematogaster sjostedti",
                                           "Crematogaster mimosae",
                                           "Crematogaster nigriceps")))
#displayed structure of nests_clean data frame
str(nests_clean)
```

```
tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ case_control   : num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ ant_species    : Factor w/ 5 levels "AB","RRB","BBR",..: 2 2 2 2 1 2 3 2 1 3 ...
 $ height_cm      : num [1:270] 392 310 513 154 324 ...
 $ scientific_name: Factor w/ 3 levels "Crematogaster sjostedti",..: 2 2 2 2 1 2 3 2 1 3 ...
```

```
#displayed 10 random rows from the data frame but made it separate from all
↪  codes above by using separate pipe operator
nests_clean |>  slice_sample(n = 10)
```

```
# A tibble: 10 x 4
   case_control ant_species height_cm scientific_name
          <dbl> <fct>           <dbl> <fct>
 1            0 RRB              142  Crematogaster mimosae
 2            0 RRB              167  Crematogaster mimosae
 3            1 RRB              220  Crematogaster mimosae
 4            0 RRB              327  Crematogaster mimosae
 5            1 RRB              204. Crematogaster mimosae
 6            0 RRB              300  Crematogaster mimosae
 7            1 BBR              216. Crematogaster nigriceps
 8            0 RRB              513  Crematogaster mimosae
 9            1 RRB              234  Crematogaster mimosae
10            0 BBR              169  Crematogaster nigriceps
```

b. Response variable

For the case_control column, the 1s mean that a tree does contain a nest and the 0s mean that a tree does not contain a nest (is an “available” tree for comparison).

c. Purpose of study

Tree height is a continuous (numeric) variable and could influence the probability of nest occurrence as if tree height were to increase, the probability of nest occurrence would increase because there would most likely be less predators higher up on trees (Results section, paragraph 2). Acacia ant species is a categorical (nominal variable) and could influence the probability of nest occurrence as if a particular ant species at a nesting site was more aggressive, the probability of nest occurrence could increase as it would reduce nest predation risks (Discussion section, paragraph 1).

d. Table of models

Model number	Tree height	Ant species	Model description
0	X	X	All predictors (no interaction term, +)
1	X	X	All predictors (interaction term, *)

e. Run the models

```
#model 0: all predictors, no interaction term
nest_model0 <- glm(case_control ~ height_cm + scientific_name,
  data = nests_clean,
#response variable is binary so test should be logistic regression
  family = binomial)

#model 1: all predictors, interaction term
nest_model1 <- glm(case_control ~ height_cm * scientific_name,
  data = nests_clean,
#response variable is binary so test should be logistic regression
  family = binomial)
```

f. Select the best model

```
#calculated AIC for each model
AICc(
  nest_model0,
  nest_model1) |>
#arranged output in order of AIC
```

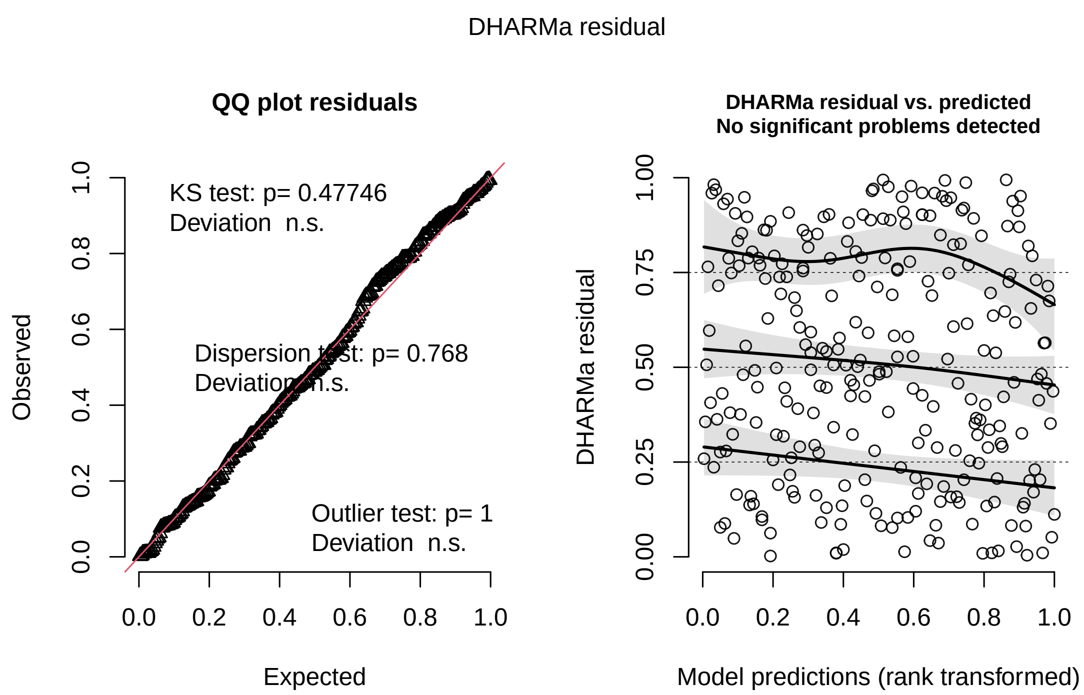
```
  arrange(AICc)
```

```
             df     AICc
nest_model1   6 238.1236
nest_model0   4 258.8940
```

The best model as determined by Akaike's Information Criterion (AIC) was the model where tree height (cm) and acacia ant species were interacting terms for the probability of nest occurrence.

g. Check the diagnostics

```
#checked diagnostics for model 1 using DHARMa residuals
plot(simulateResiduals(nest_model1))
```

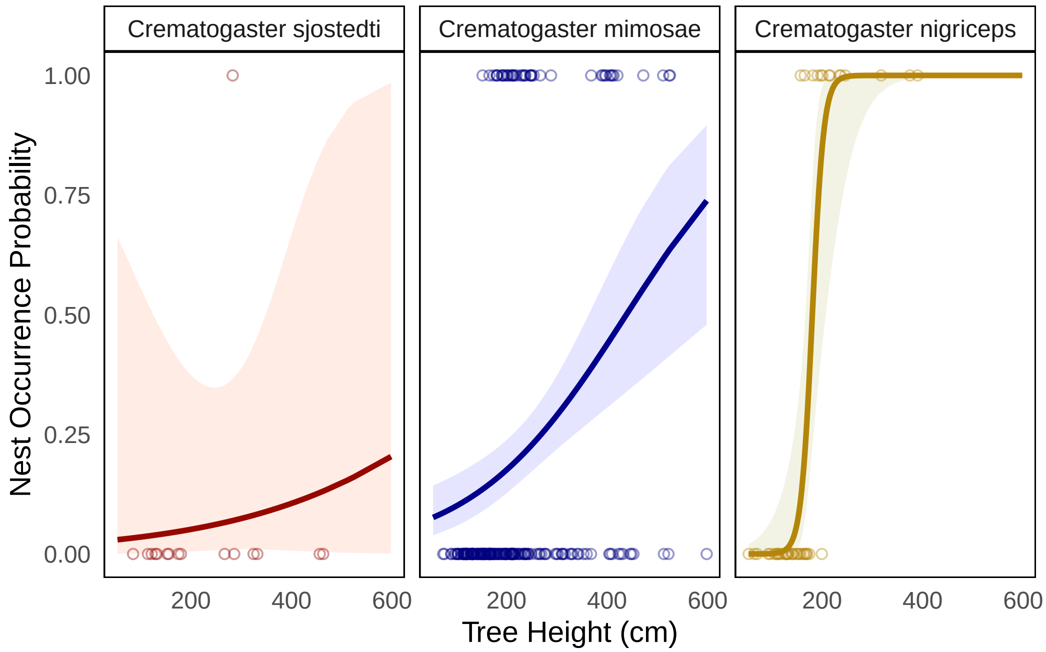


Most of the points on the QQ plot follow the reference line, so we can assume the data is roughly normally distributed. We can only believe the observations collected were independent and the response and we already know the response variable will be distributed following a binomial distribution (0 or 1). For the plot on the right, since it says there are no significant problems detected, we can also assume there is a linear relationship between transformed expected response (nest occurrence probability) in terms of link function and explanatory variables (tree height and ant species). Therefore, all of the assumptions for a generalized linear model (GLM) are met.

h. Visualize the model predictions

```
#created new object called preds
preds <- ggpredict(nest_model1,
#predictors
                   terms = c("height_cm[all]",
                             "scientific_name"))
#added scientific_name column for visualization
preds$scientific_name <- preds$group

#base layer: ggplot
ggplot(data = nests_clean,
       mapping = aes(x = height_cm,
                     y = case_control,
                     color = as.factor(scientific_name))) +
#relabelled x- and y-axes
  labs(x = "Tree Height (cm)",
       y = "Nest Occurrence Probability",
#colored by ant species
       color = "scientific_name") +
#changed colors for each ant species plot
  scale_color_manual(values = c("Crematogaster sjostedti" = "darkred",
                                "Crematogaster mimosae" = "navyblue",
                                "Crematogaster nigriceps" = "darkgoldenrod"))
                                 ↪  +
  scale_fill_manual(values = c("Crematogaster sjostedti" = "orangered",
                                "Crematogaster mimosae" = "blue",
                                "Crematogaster nigriceps" = "yellow4")) +
#used a different theme from default
  theme_minimal() +
#removed background
theme(panel.background = element_blank(),
#removed major gridlines
        panel.grid.major = element_blank(),
```

```r
#removed minor gridlines
        panel.grid.minor = element_blank(),
#removed legend
        legend.position = "none",
#added borders to each panel
        panel.border = element_rect(color = "black",
                                    fill = NA,
                                    linewidth = 0.5),
#added borders to each ant species name title
        strip.background = element_rect(color = "black",
                                        fill = "white",
                                        linewidth = 0.5)) +
#first layer: geom point
  geom_point(mapping = aes(x = height_cm, #x-axis: tree height
                           y = case_control, #y-axis: nest occurrence
                           color = scientific_name), #set to selected colors
#changed shape
             shape = 21,
#made slightly transparent
             alpha = 0.4,
#ignored global aes for colors
             inherit.aes = FALSE) +
#second layer: geom line
  geom_line(data = preds, #data from model 1 prediction
            mapping = aes(x = x, #x-axis: tree height
                          y = predicted, #y-axis: nest probability
                          color = group), #colored by ant species
            linewidth = 1) +
#third layer: CI ribbon
  geom_ribbon(data = preds, #data from model 1 prediction
              mapping = aes(x = x, #x-axis: tree height
                            ymin = conf.low, #set low value
                            ymax = conf.high, #set high value
                            fill = group), #filled by ant species
              inherit.aes = FALSE, #ignored global aes
              alpha = 0.1) + #made transparent
#made new panel for each ant species
  facet_wrap(~ scientific_name)
```



i. Write a caption for your figure

Figure 1. Probability of bird nest occurrence varies based on the acacia ant species and height of the tree. Each panel represents the data for each of the three acacia ant species, which are differentiated by colors (red: Crematogaster sjostedti, blue: Crematogaster mimosae, gold: Crematogaster nigriceps). Points represent observations of nest occurrence based on the height of the tree (cm) for each acacia ant species. Lines in each panel represent the model predictions (predicted probability) for nest occurrence based on the height of the tree (cm) and ribbons represent the 95% confidence interval for each species. Data originally from: Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. Zenodo. https://doi.org/10.5281/zenodo.8373322.

j. Calculate model predictions

```
#calculating nest occurrence from model1 for each ant species
ggpredict(nest_model1,
#predictors but set height to 600cm
  terms = c("height_cm[600]",
```

```
            "scientific_name"))
```

```
# Predicted probabilities of case_control

scientific_name: Crematogaster sjostedti

height_cm | Predicted |     95% CI
----------------------------------
      600 |      0.20 | 0.00, 0.99

scientific_name: Crematogaster mimosae

height_cm | Predicted |     95% CI
----------------------------------
      600 |      0.74 | 0.48, 0.90

scientific_name: Crematogaster nigriceps

height_cm | Predicted |     95% CI
----------------------------------
      600 |      1.00 | 1.00, 1.00
```

k. Interpret your results

If we were looking at the tallest tree, which is 600 cm, the predicted probability of nest site occupancy would be 0.2 (95% CI: [0.00, 0.99]) when Crematogaster sjostedti are present, 0.74 (95% CI: [0.48, 0.90]) when Crematogaster mimosae are present, and 1.00 (95% CI: [1.00, 1.00]) when Crematogaster nigriceps are present. In Figure 1, by looking at the lines from each panel, there seems to be a positive correlation for all three acacia ant species where as the tree height (cm) increases, the probability of a bird nest occurrence increases as well. However, this is more strongly displayed in trees that had either Crematogaster mimosae or Crematogaster nigriceps as their predicted values for nest occurrence probabilty for a tree at 600 cm were 0.74 and 1.00, respectively, which was much higher than a 600 cm tree that had Crematogaster sjostedti as its nest occurrence probability was 0.2. Therefore, acacia trees with either Crematogaster mimosae or Crematogaster nigriceps seem to be preferred by birds for nesting sites as those two ant species are a lot more aggressive defenders, reducing nest predation rates and thereby increase bird fitness in comparison to the less aggressive Crematogaster sjostedti. Furthermore, trees with Crematogaster nigriceps in particular usually have denser canopies because of the species' role in pruning apical buds, so this denser leafage makes it more appealing to birds looking to make nests in much more covered canopies.

Problem 3. Affective and exploratory visualizations

a. Comparing visualizations

- My visualizations are different from each other since for my scatterplot figure originally, I made it as smoothed out version of a histogram, which did not look right since the plot looked more like a heart rate monitor. In addition, the bar chart I made in Homework 2 was a lot more different than what it turned out to be in my affective visualization since the distribution changed between those weeks as I collected more data. Overall, the designs did change somewhat as I received feedback on how to make my histogram into a scatterplot, which I added into my affective visualization as individual points rather than a line.
- The similarities I seee between all of my visualizations are that the expected results remained relatively similar and the variables did not change at all. Looking at my visualization during the week I was doing Homework 2, I decided to continue with the same variables, only changing the histogram into a scatterplot, so I knew what the distribution was going to look like even before making my affective visualization.
- Patterns I see in my visualizations from Homework 2 are that for the bar chart, the weather I refilled my water bottle the most times in was sunny, which was expected, and for the scatterplot (technically a smoothed out histogram at the time), the frequency was a bit more random at different temperatures. However, it was different between visualizations as the in the bar chart, frequency of refilling my water bottle for cloudy and night weather conditions increased far more than windy weather by the time I got to my affective visualization. Additionally, I decided to convert the smoothed out histogram from Homework 2 into a scatterplot, so it looked a lot different on the affective visualization as it became individual points on my visual instead.
- The main type of feedback I received during week 9 in workshop was to enlarge my affective visualization in order to fill in a lot of the remaining white space of my drawing. I was able to implement this feedback by redrawing my sketch completely and for the remaining white space I did have, I colored it in with colors I felt emphasized the theme of my affective visualization. This helped a lot since I kept the foundation of my first draft sketch the same, but I refined it with different colors and designs for my figures on the affective visualization.

b. Designing an analysis

One analysis I could apply to answer my research question (How does temperature influence how often I refill my water bottle?) is a single linear regression with a Spearman rank correlation test. Since my research question only looks at one predictor and one response variable, a single linear regression is the most effective type of analysis. Additionally, the frequency I refill my water bottle is a discrete (integer/count) variable and temperature is a continuous (numeric) variable. Therefore, a Spearman rank correlation test works well as a follow-up test as it is a non-parametric test and identifies monotonic relationships between variables, which works well for discrete variables.

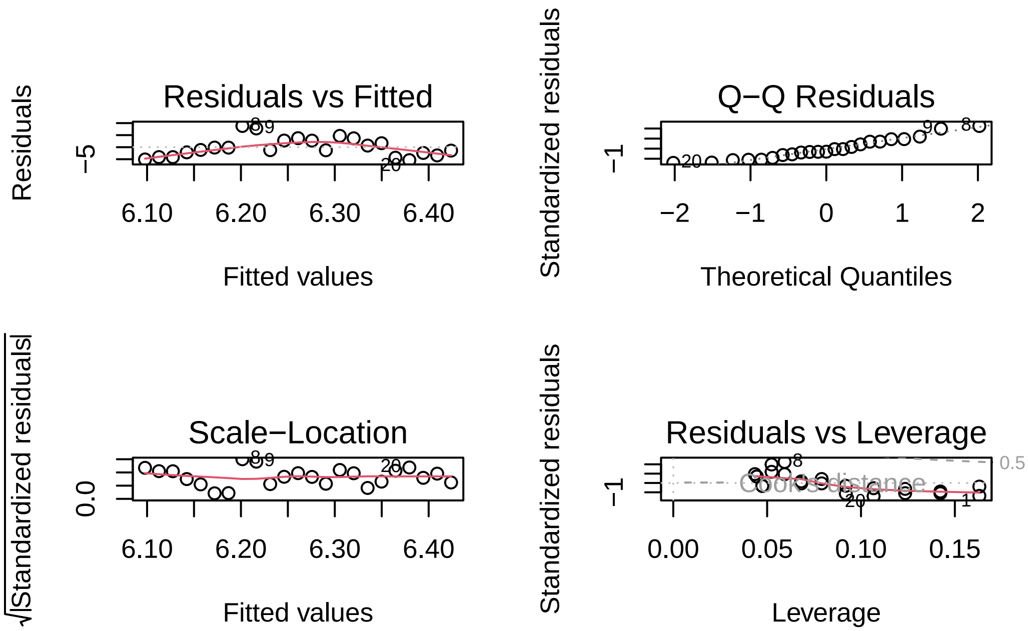
c. Check any assumptions and run your analysis

Checking assumptions.

Before fitting a model, I would expect a linear relationship between temperature and the frequency that I refill my water bottle. Additionally, I would expect independent errors, so an error from one observation I record would not influence error from another observation.

```
#created new object from personal
personal_clean <- personal |>
#cleaned column names
  clean_names()

#created a new data frame that counts the frequencies
summarized_personal <- personal_clean |>
  group_by(temperature_f) |>
  summarize(frequency = n())

personal_visual <- lm(frequency ~ temperature_f, #formula: change in refill
↪  frequency as a function of temperature
  data = summarized_personal) #data frame: summarized_personal data
```

```
#displayed plots in a 2x2 grid
par(mfrow = c(2, 2))
#displayed diagnostic plots
plot(personal_visual)
```



For the QQ plot, most of the points follow the reference line, so we can assume the data is roughly normally distributed. Furthermore, in the residuals vs. fitted and scale-location plots, the residual points are somewhat evenly and randomly distributed around the given horizontal dotted line. Therefore, all our assumptions, including the ones checked before fitting the model, have been checked for running a single linear regression model.

Running analysis.

```
#added more information about the model
summary(personal_visual)
```

```
Call:
lm(formula = frequency ~ temperature_f, data = summarized_personal)

Residuals:
   Min     1Q Median     3Q    Max
-5.379 -2.902 -1.157  2.739  8.798

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
```

```
(Intercept)    5.32708    8.03864   0.663    0.515
temperature_f  0.01482    0.12690   0.117    0.908

Residual standard error: 4.037 on 21 degrees of freedom
Multiple R-squared:  0.0006493, Adjusted R-squared:  -0.04694
F-statistic: 0.01364 on 1 and 21 DF,  p-value: 0.9081
```

```r
#ran a correlation test between temperature and refill frequency
cor.test(summarized_personal$temperature_f, summarized_personal$frequency,
#used Spearman rank correlation test as our method
         method = "spearman")
```

```

    Spearman's rank correlation rho

data:  summarized_personal$temperature_f and summarized_personal$frequency
S = 1898.4, p-value = 0.7785
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho
0.06205083
```

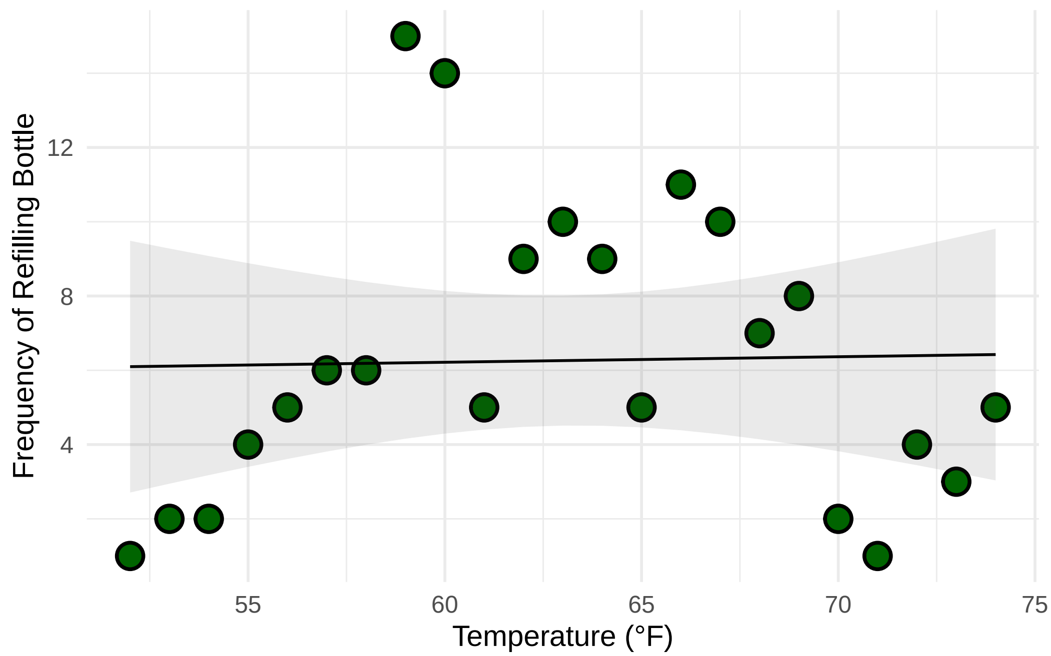
d. Create a visualization

```r
#creating a new object called abalone_preds
visual_preds <- ggpredict(
#model object
  personal_visual,
#predictor
  terms = "temperature_f")

#displayed predictions
visual_preds
```

```
# Predicted values of frequency

temperature_f | Predicted |     95% CI
--------------------------------------
           52 |      6.10 | 2.71, 9.49
```

```
         55 |       6.14 | 3.40, 8.88
         57 |       6.17 | 3.81, 8.53
         60 |       6.22 | 4.30, 8.14
         63 |       6.26 | 4.51, 8.01
         66 |       6.31 | 4.38, 8.23
         69 |       6.35 | 3.99, 8.71
         74 |       6.42 | 3.03, 9.81
```

```
#base layer: ggplot
#used summarized data frame
ggplot(data = summarized_personal,
       aes(x = temperature_f,
           y = frequency)) +
#first layer: points representing individual refills
  geom_point(size = 4,
             stroke = 1,
             fill = "darkgreen",
             shape = 21) +
#second layer: ribbon representing confidence interval
#used predictions data frame
  geom_ribbon(data = visual_preds,
              aes(x = x,
                  y = predicted,
                  ymin = conf.low,
                  ymax = conf.high),
              alpha = 0.1) +
#third layer: line representing model predictions
#used predictions data frame
  geom_line(data = visual_preds,
            aes(x = x,
                y = predicted)) +
#changed axis labels
  labs(x = "Temperature (°F)",
       y = "Frequency of Refilling Bottle") +
#changed theme from default
  theme_minimal()
```




```r
#displayed table with model coefficients, 95% confidence intervals, and more
as_flextable(personal_visual)
```

	Estimate	Standard Error	t value	Pr(>|t|)
(Intercept)	5.327	8.039	0.663	0.5147
temperature_f	0.015	0.127	0.117	0.9081

*Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '*' < 0.05*

Residual standard error: 4.037 on 21 degrees of freedom

Multiple R-squared: 0.0006493, Adjusted R-squared: -0.04694

F-statistic: 0.01364 on 21 and 1 DF, p-value: 0.9081

e. Write a caption

f. Write about your results

g. Sharing your affective visualization

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